PRODUCT USER MANUAL

For
Black Sea Wave Power

OMI_EXTREME_WAVE_BLKSEA_wave_power

Issue: 1.1

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Approval Date: November 2025

RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2024/11/26	All	Initial version	M. Ricker, A. Behrens, J. Staneva, R. Lecci	
1.1	2025/05 /28	All	Few minor revisions	M. Ricker, R. G. de Menezes , R. Lecci	Copernicus Marine Product Management

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GLOSSARY AND ABBREVIATIONS

MOI	Mercator Ocean International
NetCDF	Network Common Data Form
OMI	Ocean Monitoring Indicator
PUM	Product User Manual
QUID	Quality Information Document

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is the user manual for the CMEMS Ocean Monitoring Indicator (OMI) OMI_EXTREME_WAVE_BLKSEA_wave_power. It contains 1 OMI :

- omi_extreme_wave_blksea_wave_power

The OMI provides metrics related to recent changes of Black Sea wave power, such as 99th percentile, trend of 99th percentile, and the anomaly of the 99th percentile of the last year available in the respective multiyear dataset BLKSEA_MULTIYEAR_WAV_007_006. The OMI is based on yearly averages from 1950 to -18M in the Black Sea. In turn, the yearly averages are based on hourly instantaneous wave fields.

OMI_EXTREME_WAVE_BLKSEA_wave_power contains 2-D wave power statistics. Units are in kW m^{-1} and W m^{-1} y^{-1}.

The product is published on the CMEMS dissemination server after automatic and delayed-mode quality controls. The product is available online and disseminated through the CMEMS Information System. Files downloaded are in NetCDF format and follow CF-1. 6 convention.

Disclaimer: The quality of the product may vary during the proposed time series depending on the possible update of the system.

The OMI is described in the Quality Information Document (QUID) CMEMS-OMI-QUID-BLK-EXTREME_WAVE_BLKSEA_wave_power.pdf.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	OMI_EXTREME_WAVE_BLKSEA_wave_power
Geographical coverage	Black Sea
Variables	yearly_average_of_99th_percentile_of_wave_power 2022_anomaly_of_yearly_average_of_99th_percentile_of_wave_power yearly_trend_of_99th_percentile_of_wave_power
Available time series	N/A (provided are the overall statistics)
Temporal resolution	N/A
Horizontal resolution	1/40°
Integration depth (for example)	N/A
Format	NetCDF4

b) Production System Description

Data are extracted from the Copernicus Marine Black Sea Reanalysis dataset at 1/40°x1/40° resolution providing the significant wave height (VHM0) and wave energy period (VTM10), which are used to compute the wave power in the Black Sea. In a first step, the hourly fields are averaged to yearly metrics from 1950 to -18M . These annual metrics are then used to compute the final statistics, which are the yearly 99th percentile, 99th percentile anomaly of the last year, and the 99th percentile yearly trend.

c) Dataset details

omi_extreme_wave_blksea_wave_power contains the following variables:
P_y_avg [kW m-1**]** Yearly Average of 99th Percentile of Wave Power yearly_average_of_99th_percentile_of_wave_power
P_anom_2022 [kW m-1**]** 2022 Anomaly of Yearly Average of 99th Percentile of Wave Power 2022_anomaly_of_yearly_average_of_99th_percentile_of_wave_power
NR_y_trend [W m-1 y-1**]** Yearly Trend of 99th Percentile of Wave Power yearly_trend_of_99th_percentile_of_wave_power

3) FILE FORMAT

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link: What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article: how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

5) REFERENCES

Quality Information Document:
https://documentation.marine.copernicus.eu/QUID/CMEMS-OMI-QUID-EXTREME-WAVE-BLKSEA_wave_power.pdf
https://catalogue.marine.copernicus.eu/documents/PUM/CMEMS-BLK-PUM-007-006.pdf;
https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-BLK-QUID-007-006.pdf